Species-Aware Module Review: Sodium-Coupled Proline Uptake and Fused-PutA Catabolism in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Module:** Bacterial Na^+-coupled proline uptake + fused-PutA catabolism of L-proline → L-glutamate **Local bucket reviewed:** `kegg:ppu00250` ("Alanine, aspartate and glutamate metabolism", 36 candidate genes) **Date:** 2026-09-01

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1. Executive summary

The module is **fully satisfiable** in *P. putida* KT2440, but the local `ppu00250` candidate bucket is the **wrong container** for it and is missing its transporter. All three expected steps are encoded by just **two adjacent genes**:

- **Step 1 (Na^+/proline uptake):** `putP` / **PP_4946** (Q88D81) — a bona fide Na^+/proline symporter of the SSS family. **This gene is absent from the 36-gene candidate list** because transporters are not part of the KEGG `ppu00250` map.
- **Steps 2 + 3 (proline → glutamate):** `putA` / **PP_4947** (Q88D80) — a **1317-aa fused enzyme** with an FAD-dependent, quinone-linked proline dehydrogenase (PRODH; EC 1.5.5.2) domain and an NAD-dependent L-glutamate-γ-semialdehyde dehydrogenase (GSALDH/P5CDH; EC 1.2.1.88) domain.

Two curation-critical corrections emerge:

1. **PutA is trifunctional, not "bifunctional."** In addition to the two catalytic domains, KT2440 PutA carries an N-terminal ribbon-helix-helix (RHH) DNA-binding domain and acts as an autogenous repressor of the *put* operon. This is supported by **direct target-species structural data** (PDB **2JXI**: NMR of the *P. putida* PutA DNA-binding domain bound to its operator DNA).

2. **The module belongs to KEGG map00330 (Arginine and proline metabolism), not ppu00250.** Only 1 of the 36 candidates (`putA`) is mechanistically part of the module; ~34 are unrelated Ala/Asp/Glu/purine enzymes, and the transporter is missing. `ppu00250` should be treated as a downstream sink (PutA-derived glutamate feeds into it), not the module's home map.

Overall confidence is **high**: the assignments rest on the specific InterPro/SSS-family signature for PutP, the multi-domain architecture and pathway annotation for PutA, and a direct *P. putida* structure for the regulatory domain.

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2. Target-organism pathway definition

Included process (this module): the catabolic route by which extracellular L-proline is (i) imported across the inner membrane by a **Na^+-coupled proline symporter (PutP)** and (ii) oxidized in the cytoplasm to L-glutamate by the fused flavo-/NAD-enzyme **PutA**. PutA performs a 4-electron oxidation: the PRODH (FAD) half-reaction oxidizes proline to Δ^1-pyrroline-5-carboxylate (P5C), passing electrons to the membrane quinone pool; P5C non-enzymatically ring-opens to L-glutamate-γ-semialdehyde (GSA), which the GSALDH half-reaction oxidizes (NAD^+) to L-glutamate (P 28712849).

Alternate names / database definitions. - Enzyme names: proline dehydrogenase = proline oxidase (EC 1.5.5.2, "proline dehydrogenase (quinone)"); GSALDH = Δ^1-pyrroline-5-carboxylate dehydrogenase = P5C dehydrogenase = P5CDH (EC 1.2.1.88). - Pathway: UniProt "L-proline degradation into L-glutamate" (steps 1/2 and 2/2); KEGG map00330 "Arginine and proline metabolism"; MetaCyc "L-proline degradation" (PROUT-PWY). - Genes: `putP` (permease), `putA` (fused catabolic enzyme); the operon is the *putPA* / *put* regulon.

Neighboring pathways to keep separate. - **Proline *biosynthesis*** (opposite direction): `proB`/PP_0691, `proA`/PP_4811, `proC`/PP_5095 (+ PP_3778). These make proline from glutamate and must NOT be merged into the catabolic module. - **Osmoprotectant proline *uptake*** (different physiology): `proP`/PP_2914 (MFS osmosensory proline/betaine/H^+ permease) and the glycine-betaine/proline ABC transporter (PP_2774–PP_2775). These import proline/betaine as compatible solutes under osmotic stress; they are H^+- or ATP-coupled, not the Na^+-coupled catabolic feeder. - **Ala/Asp/Glu metabolism (`ppu00250`)**, glutamate/glutamine metabolism (`ppu00910`), arginine/urea-cycle (`ppu00220`), and 4-hydroxyproline catabolism (`proR`/PP_1258, PP_1255) — all downstream/adjacent, not part of this module.

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3. Expected step model

#	Expected step	Molecular activity	KT2440 gene	Status
1	Na^+-coupled proline uptake	proline:Na^+ symporter (GO:0005298)	**putP / PP_4946**	**covered** (add to module)
2	Quinone-linked proline oxidation	proline dehydrogenase, FAD→quinone (EC 1.5.5.2)	**putA / PP_4947** (PRODH domain)	**covered**
3	Glutamate-semialdehyde oxidation	L-glutamate-γ-semialdehyde dehydrogenase, NAD^+ (EC 1.2.1.88)	**putA / PP_4947** (GSALDH domain)	**covered**
—	(Regulatory, not in generic outline)	RHH DNA-binding autorepressor of *put* operon	**putA / PP_4947** (RHH domain)	present (module extension)

The two known inter-step relationships hold in KT2440: PutP supplies cytoplasmic proline to PutA, and PutA's P5C→GSA intermediate is passed between the two active sites. In fused PutAs this hand-off is a **substrate-channeling** tunnel connecting the ~42-Å-separated active sites, demonstrated structurally and kinetically for the PutA family (P 28712849; P 40738191; channeling first shown for a PRODH–P5CDH pair, P 25492892). Transfer of the channeling mechanism to KT2440 is **strong** (conserved fused architecture; same enzyme family and domain order).

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4. Candidate genes and evidence

High-confidence, in-module genes

putA — PP_4947 (Q88D80) — *covered (steps 2 & 3); promote to full review.* - Role: fused proline→glutamate catabolic enzyme. Domains (UniProt/InterPro): PutA RHH DNA-binding (res 11–43; IPR013321), PRODH/FAD (res ~87–567; IPR025703, IPR029041), Aldehyde-dehydrogenase/GSALDH (res 654–1101; IPR015590/IPR016161-63; active-site residues ~881/915). Length 1317 aa. - Evidence type: **direct target-species structure** for the regulatory domain (PDB 2JXI, *P. putida* PutA DNA-binding domain + operator DNA, solution NMR); UniProt catalytic/pathway annotation (EC 1.5.5.2 + EC 1.2.1.88; "L-proline degradation into L-glutamate"). Mechanistic detail transferred from the well-characterized PutA family (PMIDs 28712849, 27679491). - **Caveat:** the metadata name "Bifunctional protein PutA" and the module outline ("fused PutA … two catalytic steps") **understate** the protein — KT2440 PutA is **trifunctional** (adds autogenous transcriptional repression). This matches the report that *P. putida* PutA, unlike *P. aeruginosa* PutA, retains a regulatory function (P 12270821).

putP — PP_4946 (Q88D81) — *covered (step 1); MISSING from candidate list; promote to full review.* - Role: Na^+/proline symporter (proline permease), 542 aa. InterPro IPR011851 (Na/Pro_symporter) + IPR001734/IPR018212 (Na/solute symporter, SSS family); GO:0005298 proline:sodium symporter, GO:0031402 sodium-ion binding. Immediately adjacent to `putA`. - Evidence type: family-diagnostic signature (strong) + genomic context (adjacent to *putA*, canonical *putPA* arrangement). The *P. aeruginosa putAP* study notes 80 % PutP identity to the *P. putida* counterpart (P 12270821). PutP mechanism/Na^+-coupling is well established for the family (PMIDs 33668649, 24358297, 27793991). - **Caveat:** not in `ppu00250`; must be added to the module manually.

Candidate-list genes that are NOT in this module (representative)

All remaining 34 candidates are downstream/adjacent amino-acid or nucleotide enzymes, e.g. `glnA` / `PP_3148` / `PP_4399` / `PP_4547` (glutamine synthetases), `gltB` / `gltD` (glutamate synthase), `gdhA` / `gdhB` (glutamate dehydrogenases), `asnB`, `ansA` / `ansB`, `aspA`, `carA` / `carB`, `pyrB`, `purA` / `purB` / `purF`, `argG` / `argH`, `glmS`, `davT` / `davD`, `sad-I/II`, `gabD-II`. They handle glutamate/glutamine/aspartate/asparagine/arginine/purine/polyamine metabolism. **None** participate in proline uptake or PutA catabolism; PutA-derived glutamate merely feeds them.

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5. Gaps, ambiguities, and likely over-annotations

- **Missing transporter in metadata (gap in candidate list, not in genome):** `putP`/PP_4946 is present in the genome and is the module's uptake step but is absent from the `ppu00250` candidate set. Action: add PP_4946.
- **PutA "bifunctional" label = under-annotation:** should read trifunctional (PRODH + GSALDH + RHH autorepressor). Direct evidence: PDB 2JXI.
- **Transporter paralog ambiguity / likely over-annotation: PP_3331 (Q88HM3)** is named "Sodium:proline symporter" but its only InterPro domain is IPR005625 (PepSY-associated TM helix) — **no SSS/symporter domain**. Its proline-symporter name is a **likely over-propagated annotation**; do not count it as a second PutP without review. Mark `candidate_uncertain`.
- **Osmolyte transporters are not catabolic feeders:** `proP`/PP_2914 and the betaine/proline ABC system (PP_2774–75) import proline as an osmoprotectant; keep them out of this module (different coupling ion/energetics and physiological role).
- **Electron-acceptor detail:** PRODH is quinone-linked (EC 1.5.5.2); in aerobic *P. putida* the physiological acceptor is ubiquinone. This is inferred from EC assignment and family biochemistry (strong), not from a KT2440-specific electron-transfer measurement.
- **No direct KT2440 growth/mutant paper retrieved** in this review demonstrating proline as sole C/N source for KT2440 specifically; the phenotype is firmly established genus-wide (*P. aeruginosa* P 12270821) and mechanistically for the PutA family. Transfer to KT2440 is strong by orthology + operon structure + the *P. putida* PutA structure, but a strain-specific growth/fitness citation would upgrade this from "inferred" to "direct."

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6. Module and GO-curation recommendations

Step status calls - Step 1 (Na^+-coupled proline uptake): **covered** by PP_4946 (`putP`). *Add gene to module.* - Step 2 (quinone-linked proline oxidation): **covered** by PP_4947 PRODH domain. - Step 3 (glutamate-semialdehyde oxidation): **covered** by PP_4947 GSALDH domain. - Regulatory sub-function (not in generic outline): **present** — RHH autorepressor domain of PP_4947.

Boundary / module-document actions - `module_needs_revision` for the local bucket mapping: the module is anchored to `kegg:ppu00250`, which neither contains the transporter nor scopes proline catabolism. **Re-anchor to KEGG map00330 (Arg/Pro metabolism) / MetaCyc PROUT-PWY**, and record `ppu00250` only as the downstream glutamate sink. - Update the module's PutA descriptor from "bifunctional/fused, two catalytic steps" to **"trifunctional: PRODH + GSALDH + RHH autoregulator (autogenous *put*-operon repressor)."** Optionally add a fourth (regulatory) node or a note, since the generic outline omits it. - Keep biosynthetic (`proBAC`) and osmoprotectant-uptake (`proP`, betaine/proline ABC) genes explicitly excluded to prevent future over-merging.

GO annotations supported for KT2440 PutA (PP_4947) - GO:0004657 proline dehydrogenase activity; GO:0003842 (1-pyrroline-5-carboxylate / L-glutamate-γ-semialdehyde dehydrogenase) activity; GO:0071949 FAD binding; GO:0010133 proline catabolic process to glutamate. - Regulatory: GO:0003700 DNA-binding transcription factor activity / GO:0003677 DNA binding and GO:0045892 negative regulation of transcription — **directly supported by PDB 2JXI** (operator-bound DNA-binding domain). Recommend adding these to the KT2440 PutA record if not present. - PutP (PP_4946): GO:0005298 proline:sodium symporter activity and GO:0015824 proline transport (already annotated).

No new GO *terms* appear to be needed — existing terms cover all activities. The main deliverables are (a) adding PP_4946 to the module, (b) correcting the PutA functional description, and (c) re-anchoring the map boundary.

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7. Genes to promote to full `fetch-gene` review

1. **PP_4947** **`putA`** (Q88D80) — core catalytic + regulatory; verify trifunctional annotation, active-site residues, quinone acceptor, and add regulatory GO terms.
2. **PP_4946** **`putP`** (Q88D81) — module's uptake step, currently missing from the candidate set; confirm Na^+-coupling and add to module.

3. **PP_3331** (Q88HM3) — resolve the "Sodium:proline symporter" name vs. its PepSY-only domain content; likely over-annotation (`candidate_uncertain`).

Lower priority (boundary confirmation only): `proP`/PP_2914 and PP_2774–75 (confirm exclusion as osmolyte transporters).

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8. Key references

- Nakada Y, Nishijyo T, Itoh Y. *Divergent structure and regulatory mechanism of proline catabolic systems: … putAP operon of Pseudomonas aeruginosa PAO1 …* J Bacteriol. 2002.

  P 12270821

  (PutP ~80 % identity to *P. putida*; *P. putida*/enteric PutA retain a regulatory function.)
- PDB **2JXI** — *Solution structure of the DNA-binding domain of Pseudomonas putida Proline utilization A (PutA) bound to GTTGCA DNA* (solution NMR; maps to UniProt Q88D80/ PP_4947). Direct target-species evidence for the RHH autorepressor.
- Liu LK, Becker DF, Tanner JJ. *Structure, function, and mechanism of proline utilization A (PutA).* Arch Biochem Biophys. 2017.

  P 28712849
- Luo M, et al. (Tanner). *Structures of Proline Utilization A (PutA) Reveal the Fold and Functions of the ALDH-superfamily DUF domain.* 2016.

  P 27679491
- Sanyal N, et al. (Becker/Tanner). *First evidence for substrate channeling between proline catabolic enzymes …* J Biol Chem. 2015.

  P 25492892
- Buckley, Becker, Tanner. *Visualization of covalent intermediates and conformational states of PutA …* 2025.

  P 40738191

  (channeling tunnel; 42-Å active-site separation.)

- Henriquez T, et al. (Jung). *Prokaryotic Solute/Sodium Symporters* … 2021.
  P 33668649
  (PutP as the archetypal bacterial Na^+/proline symporter.)
- Rivera-Ordaz A, et al. (Jung). *The sodium/proline transporter PutP of Helicobacter pylori.* 2013.
  P 24358297
  (Na^+-exclusive proline symport by PutP.)
- UniProt: Q88D80 (PutA/PP_4947), Q88D81 (PutP/PP_4946), Q88HM3 (PP_3331). InterPro: IPR011851 (Na/Pro_symporter), IPR013321 (Arc RHH), IPR025703 (Bifunct_PutA). KEGG map00330 (Arginine and proline metabolism).

Evidence-strength ledger

Claim	Evidence basis	Strength for KT2440
PutP (PP_4946) is the Na^+/ proline uptake step	SSS-family InterPro signature + operon context + genus homology	Strong
PutA (PP_4947) performs both oxidation steps	UniProt catalytic + pathway annotation; conserved family mechanism	Strong
PutA is trifunctional (adds DNA-binding repressor)	**Direct** *P. putida* structure (PDB 2JXI) + P 12270821	Strong / direct
PRODH acceptor is ubiquinone	EC 1.5.5.2 + family biochemistry	Moderate (inferred)
PP_3331 is a genuine second PutP	Name only; domain content contradicts	Weak → likely over-annotation
KT2440 grows on proline as sole C/N	Genus/family evidence; no strain-specific paper retrieved here	Moderate (inferred)
